

An AI-Powered Multimedia Resource Recommendation System for Education of AI

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Abstract—This paper presents AI-Pedia, an intelligent multimedia resource recommendation system designed for the AI and machine learning (ML) education domain. The system takes user-uploaded documents (e.g., lecture notes, textbook excerpts) as input, automatically extracts key themes using TF-IDF and Maximal Marginal Relevance (MMR), and searches multiple external sources including Wikipedia, Google Scholar, arXiv, YouTube, and GitHub. A content-based filtering (CBF) pipeline ranks and recommends the most relevant text, video, and code resources. Optional AI-generated summaries enhance resource descriptions. The design targets learners who wish to reinforce their understanding with curated, domain-relevant materials. We describe the architecture, algorithms, and implementation, and outline an evaluation plan comparing the system with NotebookLM and using a “dumb student” baseline to quantify the benefit of the full pipeline. Evaluation results will be reported in a future revision.

Index Terms—Content-based filtering, educational technology, keyword extraction, machine learning, resource recommendation



1 INTRODUCTION

THIS paper presents AI-Pedia, an intelligent multimedia resource recommendation system designed for students learning artificial intelligence (AI) and machine learning (ML). Modern AI/ML courses typically combine lectures, textbooks, research papers, online tutorials, videos, and code repositories. While this abundance of material is valuable, it also creates a navigation problem: learners must decide which resources to consult, in which order, and how each item connects back to the specific concepts they are trying to understand. In practice, many students rely on ad-hoc web searches, popular video platforms, or Q&A forums. This often leads to results that are off-topic, outdated, too advanced, or inconsistent in quality.

At the same time, recent advances in large language models (LLMs) and AI-assisted tools such as Google’s NotebookLM allow users to upload documents and ask questions grounded in those materials. These tools are promising for explanation and summarisation, but they are not explicitly designed to discover external reinforcement resources across heterogeneous platforms (e.g., Wikipedia, Google Scholar, arXiv, YouTube, GitHub) or to filter those resources for a specific domain such as AI/ML. Moreover, general-purpose tools usually operate on a single interaction at a time, whereas students often need a more structured pipeline: identify key topics in their own notes, search for complementary resources, and then obtain concise summaries of what each recommended item offers.

AI-Pedia addresses this gap by building an end-to-end, content-based pipeline that starts from the learner’s own materials. Users upload a ZIP archive of AI/ML-related lecture notes, textbook excerpts, or other documents. The system automatically cleans the text, extracts representative keywords using TF-IDF and Maximal Marginal Relevance (MMR), and uses these keywords to query multi-

ple external sources. Retrieved items are filtered using an AI/ML domain keyword list and simple heuristics (e.g., de-prioritising obviously irrelevant domains) before being passed to a content-based filtering (CBF) recommender. The recommender computes cosine similarity between the user’s document representation and each candidate resource, and returns a small, ranked set of text, video, and code resources. Optional LLM-based summarisation, with a rule-based fallback when no API is available, generates short, readable descriptions for each recommended item.

The central research question of this project is: *can an automated, content-based pipeline that combines keyword extraction, multi-source search, and similarity-based ranking provide learners with high-quality, diverse reinforcement resources tailored to their uploaded documents?* This question naturally decomposes into several sub-questions: (1) how well do simple statistical methods such as TF-IDF and MMR capture the key topics in typical AI/ML course materials; (2) to what extent can multi-source search plus domain-specific filtering recover useful external resources without requiring proprietary search APIs; and (3) whether a CBF recommender built on these components can produce recommendations that learners and experts judge to be relevant, helpful, and appropriately scoped to their current level.

A secondary goal is to position AI-Pedia relative to general-purpose LLM-based tools such as NotebookLM. Both systems start from user-provided documents, but they target different usage patterns: NotebookLM focuses on conversational question answering and synthesis within a fixed corpus, whereas AI-Pedia focuses on discovering and ranking external reinforcement resources. We therefore plan to compare the two along several axes, including output quality (correctness, completeness, structure, and “teachability” of explanations), perceived usefulness for self-study,

and basic performance metrics such as response time and resource usage.

The specific objectives of this project are: (1) to design and implement a modular system for keyword extraction, multi-source resource search, and content-based recommendation focused on the AI/ML domain; (2) to integrate optional LLM summarisation while retaining a robust, purely rule-based fallback; (3) to provide a web interface for document upload, real-time progress feedback, and result download that is suitable for practical student use; and (4) to define and later complete an evaluation comprising objective comparison with NotebookLM, a “dumb student” baseline (a deliberately weakened model used to measure the added value of the full pipeline), and subjective user questionnaires. The remainder of this paper is organised as follows. Section II surveys related work. Section III describes the methodology and implementation. Section IV summarises the implemented system and its outputs. Section V outlines the evaluation plan. Section VI concludes and discusses future work.

2 RELATED WORK

This section presents a survey of existing work on the problems that this project addresses.

2.1 Main Text

Recommendation systems in education aim to suggest learning resources that match a learner’s level, goals, or current materials. Collaborative filtering uses behaviour or ratings from similar users; content-based filtering (CBF) uses features of the resources and the learner’s context to compute relevance. In our setting we use only the content of the documents the user uploads, so CBF is a natural fit [1]. Automatic keyword extraction is well-studied: TF-IDF highlights terms frequent in a document but rare in a corpus; Maximal Marginal Relevance (MMR) selects a diverse subset. Our system uses TF-IDF and MMR to choose representative keywords. General-purpose search returns mixed-quality results; for the AI/ML domain we use keyword lists and heuristics to favour authoritative sources (e.g., Wikipedia, arXiv). Our recommender uses cosine similarity on TF-IDF vectors and returns top- k resources per type. Large language models (e.g., Google’s NotebookLM) support learning from uploaded documents; our planned evaluation will compare AI-Pedia with such tools. The font used for the main text should be Palatino and the font size should be 9.5. The first line of all paragraphs should be indented by 0.5cm, except for the first paragraph of each section, subsection, subsubsection etc. (the paragraph immediately after the header) where no indentation is needed.

2.2 Figures and Tables

In general, figures and tables should not appear before they are cited. Place figure captions below the figures; place table titles above the tables. If your figure has two parts, for example, include the labels “(a)” and “(b)” as part of the artwork. Please verify that figures and tables you mention in the text actually exist. Make sure that all tables and figures are numbered as shown in Table 1 and Figure 1 below.

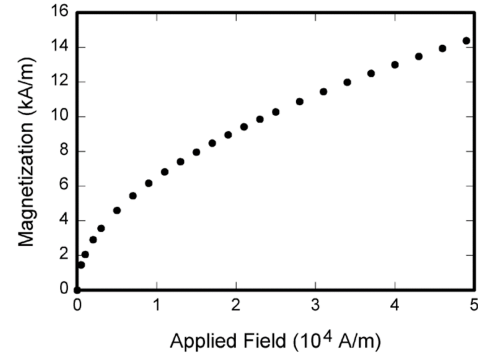


Fig. 1. Placeholder for a figure (e.g., AI-Pedia system architecture or pipeline diagram). Replace Figure1-template.png with your own image.

2.3 References

The list of cited references appears at the end of the report, listed by order of appearance. Citations use individual brackets, e.g. [1], [5]. Citation ranges are formatted as [1], [2], [3], [4] (not [1]-[4]). When citing a section in a book, give the relevant page numbers [2]. At the beginning of a sentence use author names instead of “Reference [3],” e.g. “Smith and Smith [3] show ...”

3 METHODOLOGY

This section presents the design and implementation of AI-Pedia in detail. The user uploads a ZIP of TXT or PDF documents (minimum 10). The backend extracts keywords (TF-IDF and MMR), queries multiple external sources (Wikipedia, Google Scholar, arXiv, YouTube, GitHub, etc.), cleans and normalises results, runs the CBF recommender (cosine similarity, threshold, AI-keyword filter), optionally generates summaries via an LLM or rule-based fallback, and packages outputs into a downloadable ZIP. The frontend provides upload, progress feedback via server-sent events, and download. Keyword extraction uses TF-IDF vectorisation and MMR for diversity; resource search has separate functions per type (text, video, code) with domain filtering; the recommender builds TF-IDF vectors for user documents and each resource and returns top- k per type. This section should be between 4 to 6 pages in length.

TABLE 1
Main pipeline stages of AI-Pedia

Stage	Description
1. Upload	User uploads ZIP of TXT/PDF documents (min. 10).
2. Keyword extraction	TF-IDF + MMR; output list of queries.
3. Multi-source search	Wikipedia, Scholar, arXiv, YouTube, GitHub, etc.
4. CBF ranking	Cosine similarity; threshold and AI-keyword filter.
5. Summarisation	Optional LLM or rule-based fallback.
6. Output	ZIP with txt/, video/, code/ folders.

The pipeline is modular; each stage can be configured or extended independently.

4 RESULTS

This section presents the outcomes of the implemented system. Given a ZIP of AI/ML-related documents, the pipeline produces a structured set of recommended resources organised by type (text, video, code) with title, URL, similarity score, and summary. The ZIP download contains folders `txt/`, `video/`, `code/` with cleaned content or link lists. Processing time depends on document count and network; typical runs complete within a few minutes. Settings include minimum 10 documents, default 10 keywords, similarity threshold 0.05, top-5 per type, and optional OpenAI API for summaries. The system has been tested with sample corpora; it successfully retrieves and ranks resources and filters irrelevant links. Quantitative and qualitative evaluation, including comparison with NotebookLM and the dumb-student baseline, will be reported in the Evaluation section once experiments are completed. This section should be around 2 pages in length.

5 EVALUATION

This section is reserved for the evaluation and should be between 1 to 2 pages in length. It will include: (1) objective comparison with NotebookLM (same input and tasks; quality rubrics and response time/resource usage); (2) a “dumb student” baseline (deliberately limited model vs. full pipeline; score deltas and preferences); (3) subjective user questionnaires (e.g., Likert scales on learning benefit, ease of use, trust, and preference). Results, tables, and figures will be added here in a future revision.

6 CONCLUSION

This paper presented AI-Pedia, an AI-powered multimedia resource recommendation system for machine learning education. The system takes user-uploaded documents, extracts keywords with TF-IDF and MMR, searches multiple sources (Wikipedia, Scholar, arXiv, YouTube, GitHub, etc.), and uses content-based filtering to recommend the most relevant text, video, and code resources. Optional LLM summarisation and a rule-based fallback improve the usefulness of each recommendation. The implementation is modular and configurable; a structured evaluation plan has been defined to compare the system with NotebookLM and to quantify the benefit of the full pipeline over a weakened baseline. Future work includes completing the planned evaluation, extending support to more languages and domains, and exploring tighter LLM integration. This section should be no more than 1 page in length. The page lengths given for each section are indicative and will vary from project to project but should not exceed the upper limit. A summary is shown in Table 2.

TABLE 2
Summary of Page Lengths for Sections

Section	Number of Pages
I. Introduction	2
II. Related Work	2-3
III. Methodology	4-6
IV. Results	2
V. Evaluation	1-2
VI. Conclusion	1

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